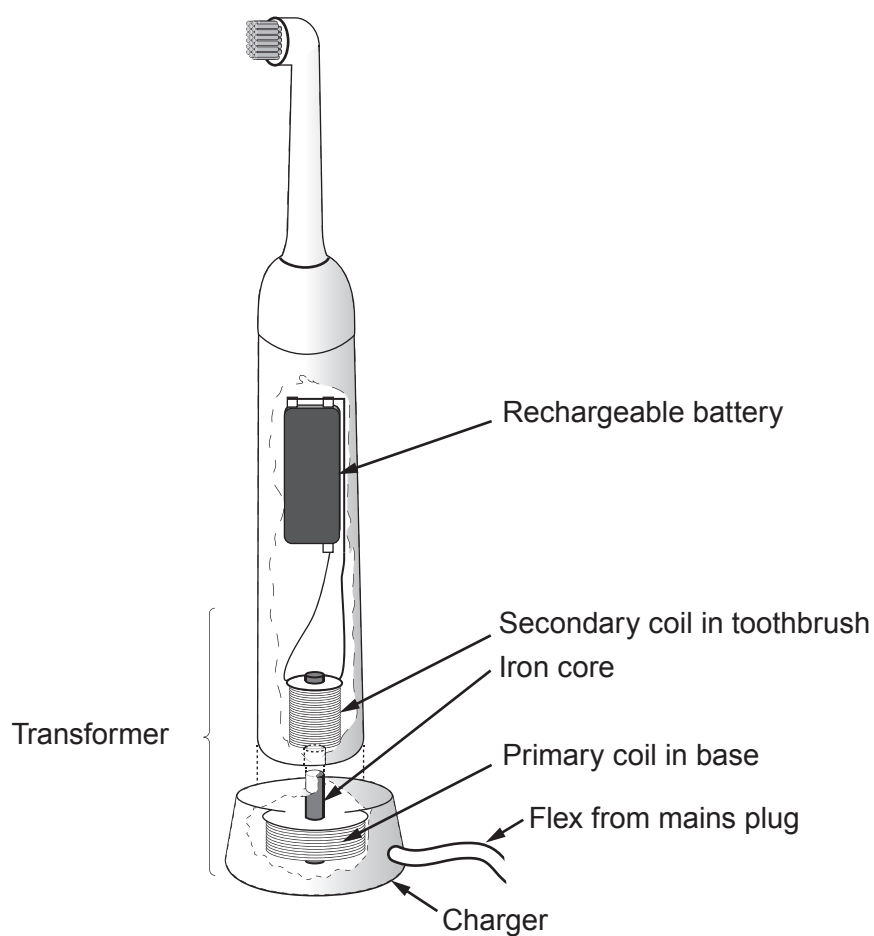


3. The diagram shows a rechargeable toothbrush which contains a transformer.



- (a) Explain how a voltage is produced in the secondary coil of the transformer. [3]

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- (b) For the following questions, assume that the transformer is 100% efficient.

- (i) Describe what is meant by *100% efficient*. [2]

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- (ii) The secondary coil inside the toothbrush has 480 turns. The primary coil in the base has 9 200 turns and is connected to the 230 V mains. Use an equation from page 2 to calculate the voltage across the secondary coil. [2]

Voltage = V

- (iii) Given that the current produced in the secondary coil is 100 mA, use an equation from page 2 to calculate the power of the toothbrush. [3]

Power = W